

DPP - 2

- Q.1 In throwing 3 dice, the probability that atleast 2 of the three numbers obtained are same is
(A) $\frac{1}{2}$ (B) $\frac{1}{3}$ (C) $\frac{4}{9}$ (D) none
- Q.2 There are 4 defective items in a lot consisting of 10 items. From this lot we select 5 items at random. The probability that there will be 2 defective items among them is
(A) $\frac{1}{2}$ (B) $\frac{2}{5}$ (C) $\frac{5}{21}$ (D) $\frac{10}{21}$
- Q.3 From a pack of 52 playing cards, face cards and tens are removed and kept aside then a card is drawn at random from the remaining cards. If
A : The event that the card drawn is an ace
H : The event that the card drawn is a heart
S : The event that the card drawn is a spade
then which of the following holds ?
(A) $9 P(A) = 4 P(H)$ (B) $P(S) = 4P(A \cap H)$
(C) $3 P(H) = 4 P(A \cup S)$ (D) $P(H) = 12 P(A \cap S)$
- Q.4 If two of the 64 squares are chosen at random on a chess board, the probability that they have a side in common is :
(A) $\frac{1}{9}$ (B) $\frac{1}{18}$ (C) $\frac{2}{7}$ (D) none
- Q.5 Two red counters, three green counters and 4 blue counters are placed in a row in random order. The probability that no two blue counters are adjacent is
(A) $\frac{7}{99}$ (B) $\frac{7}{198}$ (C) $\frac{5}{42}$ (D) none
- Q.6 South African cricket captain lost the toss of a coin 13 times out of 14. The chance of this happening was
(A) $\frac{7}{2^{13}}$ (B) $\frac{1}{2^{13}}$ (C) $\frac{13}{2^{14}}$ (D) $\frac{13}{2^{13}}$
- Q.7 There are ten prizes, five A's, three B's and two C's, placed in identical sealed envelopes for the top ten contestants in a mathematics contest. The prizes are awarded by allowing winners to select an envelope at random from those remaining. When the 8th contestant goes to select the prize, the probability that the remaining three prizes are one A, one B and one C, is
(A) $\frac{1}{4}$ (B) $\frac{1}{3}$ (C) $\frac{1}{12}$ (D) $\frac{1}{10}$
- Q.8 Of all the mappings that can be defined from the set $A : \{1, 2, 3, 4\} \rightarrow B(5, 6, 7, 8, 9)$, a mapping is randomly selected. The chance that the selected mapping is strictly monotonic, is
(A) $\frac{1}{125}$ (B) $\frac{2}{125}$ (C) $\frac{5}{4096}$ (D) $\frac{5}{2048}$

- Q.9 If m/n , in lowest terms, be the probability that a randomly chosen positive divisor of 10^{99} is an integral multiple of 10^{88} then $(m + n)$ is equal to
 (A) 634 (B) 643 (C) 632 (D) 692
- Q.10 A coin is tossed and a die is thrown. Find the probability that the outcome will be a head or a number greater than 4.
- Q.11 Let A and B be events such that $P(\bar{A}) = 4/5$, $P(B) = 1/3$, $P(A/B) = 1/6$, then
 (a) $P(A \cap B)$; (b) $P(A \cup B)$; (c) $P(B/A)$; (d) Are A and B independent?
- Q.12 If A and B are two events such that $P(A) = \frac{1}{4}$, $P(B) = \frac{1}{2}$ and $P(A \text{ and } B) = \frac{1}{8}$, find
 (i) $P(A \text{ or } B)$, (ii) $P(\text{not } A \text{ and not } B)$
- Q.13 Given two independent events A, B such that $P(A) = 0.3$, $P(B) = 0.6$. Determine
 (i) $P(A \text{ and } B)$ (ii) $P(A \text{ and not } B)$ (iii) $P(\text{not } A \text{ and } B)$
 (iv) $P(\text{neither } A \text{ nor } B)$ (v) $P(A \text{ or } B)$
- Q.14 The probabilities that a student will receive A, B, C or D grade are 0.40, 0.35, 0.15 and 0.10 respectively. Find the probability that a student will receive
 (i) not an A grade (ii) B or C grade (iii) at most C grade
- Q.15 In a single throw of three dice, determine the probability of getting
 (i) a total of 5 (ii) a total of at most 5 (iii) a total of at least 5.
- Q.16 A natural number x is randomly selected from the set of first 100 natural numbers. Find the probability that it satisfies the inequality. $x + \frac{100}{x} > 50$
- Q.17 3 students A and B and C are in a swimming race. A and B have the same probability of winning and each is twice as likely to win as C. Find the probability that B or C wins. Assume no two reach the winning point simultaneously.
- Q.18 A box contains 7 tickets, numbered from 1 to 7 inclusive. If 3 tickets are drawn from the box without replacement, one at a time, determine the probability that they are alternatively either odd-even-odd or even-odd-even.
- Q.19 5 different marbles are placed in 5 different boxes randomly. Find the probability that exactly two boxes remain empty. Given each box can hold any number of marbles.